

Week 1

1. Conducted full requirement analysis for AI-based avatar transformation service, defining scope for image editing, caching, job orchestration, and cloud storage integration.
2. Designed event-driven architecture using BullMQ as the asynchronous job processor, Redis as a broker and cache layer, and S3 as durable object storage.
3. Defined input/output queue contracts:
 - generate-ai-images
 - Update the image generation status, including lifecycle states and failure handling.
4. Created detailed request schema, including:
 - requestId
 - userId
 - avatarName
 - avatarUrl
 - attributeName
 - Category
 - Count
5. Designed an avatar caching strategy combining:
 - Redis metadata (TTL-based)
 - Local filesystem persistence to reduce S3 bandwidth and latency.
6. Designed job reliability mechanisms:
 - Explicit acknowledgment
 - Lock duration tuning
 - Controlled concurrency
 - Retry handling strategy
7. Produced architecture diagrams, request flow documentation, and technical design blueprint for the implementation phase.